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Research Commentary

Data-Driven Computationally Intensive Theory Development

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Abstract. Increasingly abundant trace data provide an opportunity for information systems researchers to generate new theory. In this research commentary, we draw on the largely "manual" tradition of the grounded theory methodology and the highly "automated" process of computational theory discovery in the sciences to develop a general approach to computationally intensive theory development from trace data. This approach involves the iterative application of four general processes: sampling, synchronic analysis, lexical framing, and diachronic analysis. We provide examples from recent research in information systems.

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Keywords: grounded theory methodology • computational theory discovery • GTM • computational • trace data • theory development • lexicon • inductive

1. Introduction

The abundant and ever-increasing digital trace data now widely available offer boundless opportunities for a computationally intensive social science (DiMaggio 2015, Lazer et al. 2009). By "trace data" we refer to the digital records of activity and events that involve information technologies (Howison et al. 2011). Given the ubiquitous digitization of so many phenomena, some expect the widespread availability of a variety of trace data to do nothing less than revolutionize the social sciences and challenge established paradigms (Lazer et al. 2009). Through direct computational attention to trace data, researchers can generate richer and more accurate understandings of social life—insights closer to the source (Latour 2010). Trace data typically require computational tools for novel visualizations and pattern identification (Lazer et al. 2009), which provides ample fodder for predictive modeling (Shmueli and Koppius 2011). But such models, patterns, and visualizations are not *theory* (Agarwal and Dhar 2014).

To unleash the power of trace data, information systems researchers can benefit from a general approach—a common ground across perspectives—for inductively generating novel theory from these data in all their forms. In this research commentary, we describe such an approach, rooted in the grounded theory method (GTM; Glaser and Strauss 1967), yet also informed by computational theory discovery (CTD) in science fields. We propose a general, computationally intensive approach to the inductive generation of theory from trace data. By describing the approach as "computationally intensive," we seek to emphasize that it is neither classic, manual GTM nor entirely automated CTD. Instead, there is a combination of manual and automated activity. The process involves key activities—sampling, synchronic analysis, lexical framing, diachronic analysis—and builds on the key idea of emergence through iteration across these activities. We highlight the important role of the theoretical and "pre-theoretical" vocabulary, or lexicon, within which researchers frame the trace data to construct theory. Although the importance of a sensemaking lexicon may seem obvious, it is important to appreciate the theoretically loaded character of scholarly lexicons when generating theory from trace data. The choice of lexicon matters; it both enables and constrains the theoretical contribution that one can construct from trace data.

We thus look to extend the principles and spirit of GTM for alternative empirically grounded inductive approaches that do not necessarily follow the prescriptions of GTM. This can perhaps make way for a new generation of methodological prescriptions specifically suited to computationally intensive analysis of trace data and their combination with more traditional forms of GTM analysis. Particularly in the context of widely available trace data and computational social science, the unprecedented access to different forms of data can drive novel inductive approaches that are consistent with the general approach of GTM, but perhaps not with existing, established methodological guidance.

We proceed as follows. In the next section, we define trace data and provide an overview. Then we highlight the role of lexicons in enabling and constraining theory development, and we compare "manual"

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grounded theory development and the "automated" process of computational theory discovery. Grounded in this analysis, we develop a general approach to computationally intensive theory development. Our resulting framework is intended to guide empirically grounded theory construction based on any kind of data using a variety of automated and manual techniques. We illustrate our approach with three published cases, and we conclude by reviewing the contributions of this commentary.

2. Trace Data, Grounded Theory, and Computational Theory Discovery

When an activity or event occurs in conjunction with information technology (IT), it often leaves a digital record, or "trace." The term "trace data" refers to digital records of activities and events that involve information technologies. Trace data constitute a form of unobtrusive measure (Webb et al. 1966) that is enabled by digital technologies. Trace data are different from many other common forms of social science data in a variety of ways (Howison et al. 2011). Often, trace data are "found" data—a byproduct of activities, not data generated for the purpose of research. In the case of qualitative data, for example, analysis of existing texts would involve trace data, whereas analysis of interview transcripts conducted for the research project would not. Furthermore, trace data are "event-based" records of activities and transactions. Therefore, trace data are longitudinal and can take the form of time-stamped sequences of activities. Clickstreams, sensor data, and social media updates are all time-stamped sequenced trace data, but a cross-sectional record of, say, user attitudes or intentions is not.

By our definition, information systems researchers have been analyzing various forms of trace data for decades. Texts such as emails or other documents, transaction data from organizational systems, and social media updates are all forms of trace data. One might even conclude that the information systems field is particularly well suited to the study of trace data (Agarwal and Dhar 2014, Howison et al. 2011). What is new, however, are the ever-increasing aspects of virtually every phenomenon that now leaves digital traces, and these are expected only to increase further (Lazer et al. 2009). A few decades ago, trace data involved only data that were stored in a handful of organizational systems. Now the number and breadth of organizational systems has increased dramatically. In the past, a good deal of organizational activity occurred outside the purview of organizational systems. Given the widespread adoption of enterprise information systems, document and content management systems, advanced productivity applications, and others systems, most organizational activities now leave some sort of trace in terms of log files and communication or document trails. Furthermore, devices are more abundant in organizational activity, including mobile phones, specialized mobile devices, various sensor and tracking technologies, and elements of the emerging Internet of Things. Outside of the organization, more and more people are using social media, mobile applications, and an ever-increasing number of sensors associated with the "digitized self"—homes, cities, and societies are all becoming sensitized.

Given this abundance of data, researchers can investigate a multitude of questions, increasing the number and variety of researchers who will investigate information systems phenomena through trace data. Certainly, traditional hypothetico-deductive (hypothesis testing) methods will continue to be dominant approaches to analyzing trace data, but the temptation to engage in open-ended exploration of these abundant data will also be strong to gain insight into a variety of phenomena. As information systems researchers look to generate theory from trace data, a common ground can be helpful to communicate across traditions.

3. Manual vs. Automated Data-Driven Theory Development and the Role of the Lexicon

To develop a general approach to theory development from trace data, we highlight the importance of the researchers' lexicon in enabling and constraining a theoretical contribution, and we compare two polar traditions in inductive theory development—the intensely manual, largely qualitative tradition of grounded theory methodology and the highly automated, quantitative process of computational theory discovery. In doing so, we look for commonalities (see Figure 1).

3.1. The Role of Lexicon in Developing Theory Grounded in Data

The language choices that researchers make are fundamental to any scientific endeavor and critically important for eliminating ambiguity in research and enabling research traditions to move forward (Podsakoff et al. 2016). In his seminal work on organizational theory, Bacharach (1989) points out that "theory" is essentially a linguistic device that researchers use to organize empirical data in a way that simplifies those complex data with the use of concepts, and that asserts certain relationships among those concepts within some boundary condition and constraints.[1] Researchers construct theories through iterative, creative reasoning, but this theorizing does not occur from a blank slate—researchers necessarily draw upon prior scholarship in the theory construction process (Van de Ven 2007).

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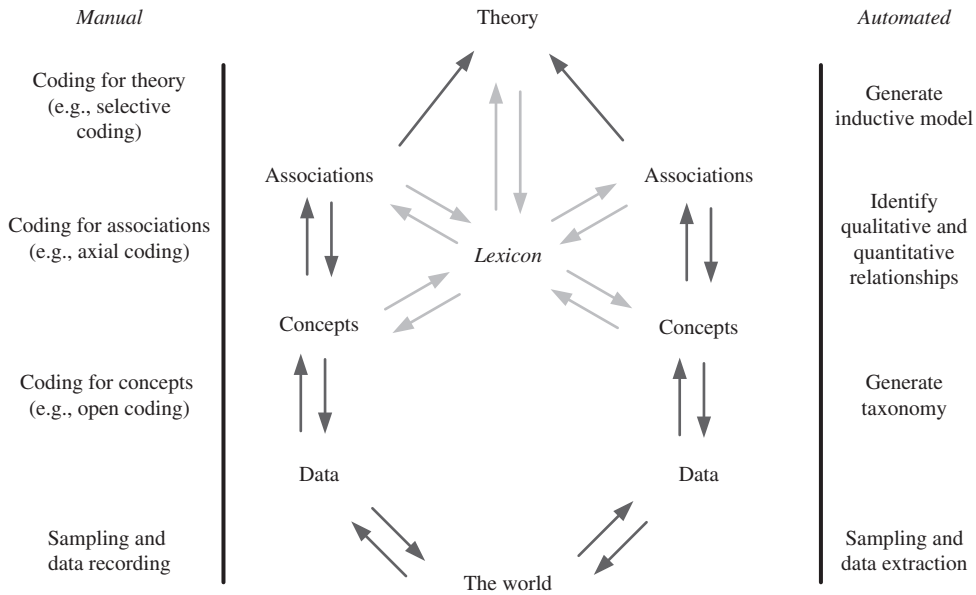
Figure 1. The Role of the Lexicon in Manual and Automated Empirically Driven Theory Generation



Note. Black arrows describe the iterative process of building theory, with the general direction going from empirical to theoretical; gray arrows describe referencing to and from the lexicon in this process.

In his philosophy of scientific knowledge, Habermas (1984, 2003) pointed out that different communities of researchers use language in very specific, theoretically loaded ways. When analyzing empirical data through a theoretical lens, scientists use a lexicon shared by their community, which provides ready-made constructs and statements of relationships that they can then build on. Habermas (quoted in Pedersen 2008, p. 462) referred to a lexicon as the "pre-theoretical" grammar that is required for building any theoretical contribution. Situating scientific work in a lexicon both enables and constrains the scientific contribution. The lexicon enables because researchers do not have to reinvent all theoretical relationships from the ground up, and the lexicon acts as a pre-theoretical basis for their contribution. The lexicon constrains the contribution because, in choosing a particular lexicon, scientists adopt the path-dependent foundation that limits the degrees of freedom for their theoretical contribution.

Researchers do not create the entirety of their language anew each time they theorize. Instead, they use the pre-existing set of concepts, constructs, and their implied relationships as a starting point. That is, language is always generated in relation to preexisting theoretically loaded language. Thus, any new theorizing necessarily draws upon the lexicon of a particular scientific community. The lexicon is not a trivial issue of word choice, but in the scientific endeavor is critical to enabling and constraining any contribution to knowledge. Next we describe the process of manual grounded theory development and then compare it with computational theory discovery, highlighting the role of the lexicon in each.

3.2. The Process of Manual Grounded Theory Methodology

Grounded theory methodology (Glaser and Strauss 1967) has been one of the strongest catalysts for widespread acceptance of qualitative research as well as inductive theory building across a variety of social science disciplines (Bryant and Charmaz 2007, Eisenhardt 1989). Grounded theory seeks to develop theoretical concepts and relationships while being informed by intense analysis of empirical data (Glaser and Strauss 1967, Strauss and Corbin 1990).

Over the years, GTM has evolved into a contested "family" of methodologies, rather than one very specific method (Bryant and Charmaz 2007). This family of methods is replete with variants and rich in reflective discourse (Walsh et al. 2015). There are disagreements on coding procedures (e.g., Kelle 2007), the role of existing research (e.g., Jones and Noble 2007), epistemological foundations (e.g., Charmaz 2000), and a host of other divisions (also compare Seidel and Urquhart 2013). From a unifying perspective, however, the method can be thought to involve building or extending a lexicon in a substantive area of investigation while, at the same time, drawing on an existing lexicon as pre-theoretical understanding in support of further sensemaking with observations in the data.

Traditional manual GTM begins with the world's biggest data set—the world itself—and reduces this data set by sampling from the world in an area of interest. This sampling should be theoretical—what is known as "theoretical sampling"—in that the sample should be developed and extended based on the results of analyzing that existing sample (Glaser and Strauss

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1967). In this view, a smaller, initial sample should be taken and analyzed, then subsequent samples should be informed by this analysis—they should help to follow up on the insights that began to emerge from the initial sample. As such, the sample emerges over time, and this emergence is informed by existing analysis.

Coding and categorizing data are fundamental activities in GTM, and many of the prescriptions for qualitative coding involve an intensely manual process (Charmaz 2006, Goulding 2002, Holton 2007). These coding strategies may transfer directly to trace data, such as "trace ethnographies" (Geiger and Ribes 2011, p. 1) or "discourse archives" (Levina and Vaast 2016, p. 216), and some coding processes can likely be automated with machine learning, natural language processing, and other computationally intensive techniques. Coding is not necessarily for only qualitative data; it can also apply to quantitative data (Glaser 2008).

The process of coding involves multiple passes through the data, iteratively identifying concepts and categories (i.e., more abstract concepts) that become more general at each pass, and then iteratively relating these concepts and categories to each other, resulting in the generation of theory. In the spirit of theoretical sampling, this analysis informs additional data collection, which then informs subsequent rounds of analysis—much the way a detective follows up on new leads given new information (Morse 2007). This continued sampling and analysis may involve various qualitative and quantitative data sources, including interviews (i.e., the coding of someone else's statements), observations such as online community threads, or memos written by the researcher throughout the analysis (Levina and Vaast 2016). In GTM, coding is not something that happens after the data are collected, but occurs in interaction with the data collection; each informs the other, and coding is shaped by and, in turn, shapes different approaches to data collection. Codes reflect the constant comparison of emergent analysis with existing bodies of knowledge and their respective lexicons. Thus, there cannot be any grounded theory development without a pre-theoretical lexicon, and the myth of the researcher as a "blank slate" has been repeatedly debunked (Urquhart and Fernández 2013). While the pre-theoretical lexicon is not necessarily applied in the sense of preconceived, a priori concepts and relationships, it is drawn upon over the course of the research by the analyst to enhance her theoretical sensitivity in interactions with the field (Charmaz 2006).

This process (see the left side of Figure 1) of manual GTM can be summarized in the following steps:

(1) *initial sampling* from the world, then continued rounds of *theoretical sampling*, to record data;

(2) iterative coding to *identify concepts*, drawing on one or more lexicons;

(3) further coding and pattern matching to *identify associations* and relationships, again drawing on the salient lexicons;

(4) iterative sensemaking of associations in relation to the pre-theoretical and theoretical understandings of existing lexicons in the relevant fields to *construct theory*.

The data sample, the concepts and associations, the lexicon, and the resulting theory emerge from an intensely iterative process over time. Through coding and analyzing the data, the analyst moves from the descriptive to the conceptual level, and the results of this process are statements of the relationships between concepts (Holton 2007) that together constitute theory. This analysis process involves both synchronic (i.e., identification of concepts and associations in any given moment in time) and diachronic (i.e., identification of time-dependent relationships between concepts, for instance, in terms of cause–effect relationships) approaches to analyzing data (Holland et al. 1986). Coding and analysis can follow a number of paths (Charmaz 2006; Glaser 1978, 1992; Strauss 1987; Strauss and Corbin 1990, 1998; Urquhart 2013), the most well known of which are the open, axial, and selective coding cycles in Straussian GTM (e.g., Strauss and Corbin 1990, 1998). While there has been intensive debate about—and disagreement on—the different coding strategies proposed in different approaches to GTM[2] (e.g., Bryant and Charmaz 2007, Duchscher and Morgan 2004, Matavire and Brown 2011), all versions of grounded theory involve the four stages of sampling, identification of concepts, identification of associations, and the construction of an integrated theoretical scheme. This coding process is fundamental to GTM and is generally a manually intensive process.

3.3. The Process of Automated Computational Theory Discovery

On a general level, the process of the grounded theory methodology has striking parallels to computational theory discovery—a discipline that emerged in the 1970s to automate the process of scientific research in the hard sciences to produce "discoveries" through artificial intelligence and machine learning techniques (Džeroski et al. 2007). These techniques accompany a worldview that sees hypothetico-deductive methods as artifacts of computational limits that might be outdated remnants of history:

> While the history of science can serve as an argument for norms of practice, for several reasons it is not a very good argument. The historical success of researchers working without computers, search algorithms, and modern measurement techniques has no rational bearing at all on whether such methods are optimal, or even

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> feasible, for researchers working today. It certainly says nothing about the rationality of alternative methods of inquiry. Neither *was* nor *is* implies *ought*. The "Popperian" method of trial and error dominated science from the sixteenth through the twentieth century not because the method was ideal, but because of human limitations, including limitations in our ability to compute. (Glymour 2004, pp. 75–76)

The history of science is rife with examples of discovery of theories from observations, a process that modern epistemologists and scientists sought to understand. Herbert Simon (1977) proposed a view of theory discovery as heuristic problem solving. In this paradigm, scientists use mental operators to advance through a large search space from one knowledge state into another. Newell drew on this idea to provide a framework for both a theory of human problem solving and an approach to building computer programs with similar capabilities (Džeroski et al. 2007). Computational theory discovery is rooted in this view and seeks to explicate human intellect and construct a computational implementation of discovery processes (Wagman 1997). Computational theory discovery approaches have enjoyed successful outcomes in a variety of scientific fields, including in mathematics, physics, chemistry, and genetics (Wagman 2000).

In addition to CTD, various computational techniques for learning generalizable models from observations were developed in the discipline of machine learning. In particular, *knowledge discovery in databases* (KDD) emerged in the 1990s to make sense of large transactional data sets by mapping low-level, hard-to-understand data to higher-level concepts (Fayyad et al. 1996). Recently, KDD received acceptability in the scientific community (Gaber 2010). KDD and CTD share the same premise of using data to extract patterns and identify hypotheses (Williamson 2009). Indeed, pioneers of the two disciplines pointed to their commonalities (Klösgen and Żytkow 1996).

These computational disciplines share an underlying inductive framework and they are all geared toward extracting patterns from data and learning higher-level models and representations (Glymour et al. 1996). Across a variety of fields "econometricians, statisticians, and data mining specialists are generally looking for insights that can be extracted from the data" (Varian 2014, p. 5). In KDD, the progression from data to knowledge proceeds throughout the five steps of selection, preprocessing, transformation, data mining, and interpretation and evaluation (Fayyad et al. 1996, p. 41). In CTD, Langley's (2000) model outlines four major steps, rooted in the three types of scientific knowledge that constitute the major products of the scientific enterprise (Džeroski et al. 2007). Following Langley's (2000) model, the process of theory discovery (see the right side of Figure 1) can be summarized in the following steps aiming at generating scientific knowledge from observations:

(1) *sample observations* from phenomena of interest;

(2) iteratively *generate a taxonomy* of concepts from observations, drawing on one or more lexicons;

(3) *identify qualitative and quantitative relationships* and associations among concepts of the taxonomy;

(4) iteratively *generate structural and process models* by drawing on associations in relation to the pretheoretical and theoretical understandings of existing lexicons in the relevant fields.

It is important to note that computational theory discovery is automated, but not automatic. Concepts are organized around people's theories about the world—the background knowledge and lexicons that guide and constrain learning (Wisniewski and Medin 1991). There is a significant element of human interaction in all stages of the process. In particular, the role of humans is critical in the sampling process—in choosing which data to analyze and why. Humans choose a data sample to address a particular problem, and this problem formulation is a key element of any such analysis, and is inevitably an intensely human endeavor (Simon 1977). Humans interact with automating systems throughout the process—which often includes additional data collection and validation processes. Without intense human interaction, CTD projects can readily fail (Gaber 2010). Computational theory discovery is not intended to supplant the role of the researcher, but to amplify it (Glymour 2004, p. 77). Human knowledge has been shown to be superior to machine learning in identifying problematic cases for which data are scarce, and therefore the role of a human in even the most automated process cannot be underestimated (Attenberg et al. 2015).

4. A Computationally Intensive Approach to Theory Development

In this section, grounded in our analysis of both manual and automated theory discovery, we develop an approach that allows for different computationally intensive grounded theory techniques ranging from predominantly manual theory development to predominantly automated theory discovery. Such abstraction is consistent with the idea of grounded theory as a meta theory allowing for all sorts of instantiations, where analysts combine different manual and automated methods (Walsh 2015), with varying degrees of computational intensity.

A key insight from our analysis of these two approaches to inductive theory generation is that *all* research nowadays involves both manual and computational components. In the time that Glaser and Strauss (1967) conducted their path breaking research, the process of manual grounded theory generation

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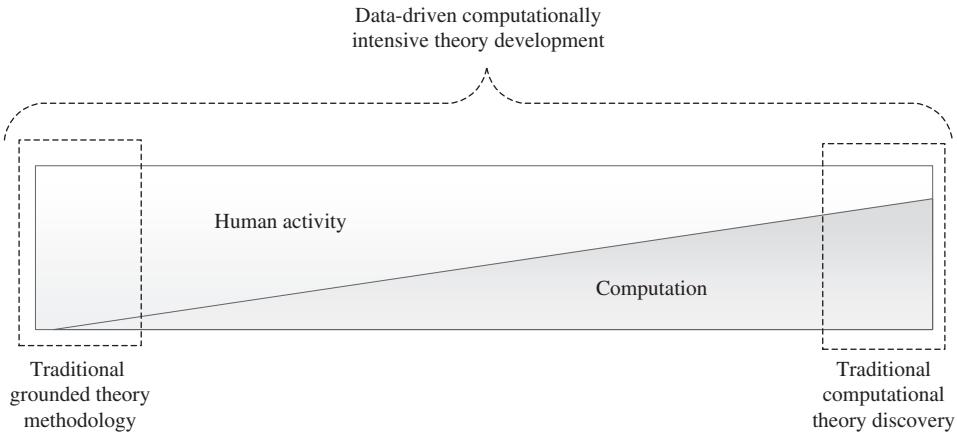
Figure 2. Data-Driven Computationally Intensive Theory Development—Combining Human and Computational Methods in Varying Proportions



Notes. Note that the image is not symmetrical, to indicate that, although there can be a purely manual grounded theory approach, there is no entirely computational approach to theory discovery in that it inevitably involves human activity (the right side of the figure). The relative magnitude between these poles is for illustrative purposes only and is not intended to illustrate the relative space of one approach versus another in any way.

took place primarily (if not entirely) on paper—but this is no longer the case. Researchers transcribe, code, and analyze their data using all sorts of qualitative and quantitative software tools. Nevertheless, we refer to this approach as "manual" to compare it with computational theory discovery techniques. Similarly, computational techniques are clearly not entirely automated, but inevitably involve manual guidance and human judgment (Todorovski and Džeroski 2007). Between these two approaches, there is a space for a host of combined, "computationally intensive" techniques that offer fertile opportunity for theory generation (Figure 2).

Building on the two poles of existing traditions for inductive theory generation—GTM and CTD—we can now propose an abstracted process for combined, computationally intensive grounded analysis, focusing on the role of the lexicon in enabling the generation of theory from patterns identified in the data. Table 1 summarizes the main activities for both manual grounded and automatic computational theory generating processes—integrating the two approaches to highlight the potential for their interplay in a study.

It is important to emphasize that manual analysis and computational analysis are complements rather than substitutes. The researcher adopting a computationally intensive grounded theory approach can integrate manual and computational analyses. For example, when sampling and collecting data, one might start with theoretical sampling via interviews and later identify opportunities to enrich the data set with trace data—or vice versa. In synchronic analysis, the researcher can classify trace data using codes identified manually or identify such codes computationally using clustering and validate them manually. Associations uncovered computationally are manually assessed for content validity. Manual associations between codes can benefit from a rigorous computational treatment. In diachronic analysis, the researcher may validate and quantify theories that were manually grounded or make sense of structural and process models that were computationally discovered. Theorizing is a process of sensemaking and abstraction that demands human ingenuity and creativity. Computational methods can increase the efficiency and reliability of researchers by allowing them to examine vast quantities of data and consider various questions that can arise from the data simultaneously (Glymour 2004, p. 77). Furthermore, it is important to note that these activities do not follow a sequence in discrete steps, but iterate and emerge across the steps as the exploratory research unfolds. In the following, we provide a summary of each iterative step.

4.1. Sampling and Data Collection

At the outset, the analyst defines the area of investigation, thereby defining the scope and boundary conditions of the intended theory development. Often this begins by convenience (a data set or study location is available) or by a phenomenon (some topic domain is "hot" at some point, so researchers look to explore that domain). In early stages of research, an initial sample is drawn and analyzed. While in manual grounded theory the researcher often actively contributes to the process of data collection (e.g., through interviewing), trace data are typically found data (e.g., generated through user activity). The process of further sampling guided by insights from this first round of analysis ensues (Glaser and Strauss 1967). In traditional grounded theory methodology, the sampling process

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Table 1. Combining Activities and Goals in Manual and Automated Analysis

Activity	Goal	Manual grounded theory methodology	Automated computational theory discovery
Sampling and data collection	Iteratively develop data set	Theoretical sampling	Recording observations
		Combination: Iteratively constructing a data set through cycles of data collection as a result of interaction with the phenomena of interest and the digital traces of those phenomena.	
Synchronic analysis	Categorize the data using concepts and identify associations among concepts	Coding for concepts and associations (e.g., open coding and axial coding in Straussian GTM)	Create taxonomy (e.g., using cluster analysis or association rule mining)
		Combination: Iteratively categorizing data according to established concepts and looking for qualitative and quantitative relationships and associations of these concepts to each other in the data.	
Lexical framing	Draw upon and extend the language of one or more research communities	Codes and relationships	Taxonomy and associations
		Combination: The lexicon provides the pre-theoretical reference for the naming of concepts and the identification of patterns in relation to a goal, using the language and causal relations determined by one or more scholarly communities.	
Diachronic analysis	Generate theory	Constructing theory (e.g., selective coding in Straussian GTM) Theorizing (e.g., temporal bracketing, grounded process theorizing)	"Develop model" (e.g., correlations, regressions and decision trees) Develop inductive model (e.g., process induction, process mining)
		Combination: The generation of theory requires a sensemaking process. Rooted in empirical evidence, the analyst decides what concepts and relationships (pre-theoretical understanding) to include in elaborating a coherent theoretical scheme (theoretical understanding) and thus proposing an extension to the knowledge of a particular community of researchers.	

(theoretical sampling) is expected to be very focused, in part because of the cognitive limits of individuals. For example, pointing out the need for efficient sampling, Morse (2007, p. 233) stated:

> Computer programs, while invaluable, merely assist in placing the data in the best possible position to aid the researcher's cognitive work; such programs cannot actually do the analysis for the researcher. It is for this reason that collecting too much data results in a state of conceptual blindness on the part of the investigator. Excessive data is an impediment to [GTM] analysis, and the investigator will be swamped, scanning, rather than cognitively processing, the vast number of transcripts, unable to see the forest for the trees, or even the trees for the forest, for that matter.

This sentiment is probably quite accurate for a strictly manual approach for grounded theory. Individuals pouring over qualitative data need to do so in part by minimizing the data set in the interests of efficiency. However, the moment one recognizes the analytic benefits of computational technologies, one can appreciate how an interplay of analyses between qualitative and trace data provides better opportunity for theorizing. The spirit of theoretical sampling remains—one begins with a convenience sample, and this sample can be intentional (like conducting interviews) or can involve inductive analysis of trace data. Based on initial findings, the researcher then samples additional data, either of the same type or of a complementary type. According to Gaskin et al. (2014, p. 851), this mixed analysis of qualitative and computational data (for example) enables researchers to "zoom in and out" of phenomena—zooming in to get a rich understanding of elements of the data in context, and zooming out to look for and verify broader patterns. Combining sorts of data in the iterative sampling process helps researchers avoid merely "rationalizing" (Garud 2015) what they see in terms of a particular perspective.

4.2. Synchronic Analysis

In conjunction with the rounds of sampling, researchers continually explore the data. In manual grounded theory, this involves coding for both concepts and associations between concepts, and in computational analysis, this involves developing a taxonomy. Holland

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et al. (1986) describe the categorization and raw association of concepts in terms of *synchronic* regularities. In the earlier stages of manual grounded theory, the researcher aims to identify first categories based on the similarities between empirical indicators as well as first co-occurrences of categories. In open coding (Strauss and Corbin 1998), for instance, the analyst identifies categories by grouping similar incidents found in the data under the same label. In axial coding (Strauss and Corbin 1998), the analyst looks for other categories (subcategories) that co-occur with this category;[3] that is, the analyst looks for both similarities (grouping) and correlations (co-occurrence of categories and their subcategories). In the computational analysis of trace data, both processes (identifying categories and identifying associations) involve a process of converging to synchronic relations using a variety of clustering techniques (Duda et al. 2001, Hastie et al. 2003). Clustering associates observations in data to clusters based on their similarity. Observations in the same clusters share recurrent patterns or synchronic associations. Very often, the challenge for constructing synchronic associations is the exponential number of such relationships (Glymour et al. 1996, p. 39). CTD focuses on finding parsimonious, understandable, and communicable sets of relationships (Schwabacher and Langley 2001). Identifying synchronic relations need not be either qualitative or computational, but can be both (Anderberg 1973, Hipp et al. 2000).

4.3. Lexical Framing

Iterating with the coding and continued sampling (as necessary), the analyst settles upon the lexical frameworks to be used to analyze the data—that is, the pre-theoretical lexicon providing an appropriate grammar for analyzing the data. We refer to this conscious activity of drawing on a lexicon to attribute meaning to codes as "lexical framing" (Fillmore 1976, p. 27). This can, of course, involve drawing on multiple lexicons, but typically addresses that of the focal community of researchers, or "conversants" (Huff 1999). In the analysis process, the researcher may consider different pre-theoretical lexicons throughout the process, and the lexicon in use may change. Furthermore, certain lexicons may be more or less abstract, which can influence the scope of emergent theorizing. Different levels of abstraction can be combined, as well. For example, one can draw upon abstract pre-theoretical lexicons such as the *coding paradigm* including labels such as "conditions," "actions/interactions," and "consequences" (Strauss and Corbin 1998), and combine this with very specific classification schemes for computational analysis, like labeled or curated data sets for training learning algorithms.

Similarly, one can explore a data set using multiple forms of cluster analysis techniques (Anderberg 1973), but use *conceptual clustering* to supervise the algorithm by drawing on specific theoretical discourse, allowing the researcher to incorporate the desired aspects of categories that are independent of the data (Fisher 1987, Michalski 1980). Conceptual clustering involves using known attributes to categorize data and mimics human concept learning, where concept formation relies on prior knowledge (Thompson and Langley 1991).

4.4. Diachronic Analysis

Since theory involves identifying causal or sequential patterns, theory development implicitly or explicitly involves a temporal element—and therefore theory construction necessarily rests on diachronic, temporal analyses (Holland et al. 1986). Even in trace data, which necessarily have a temporal character, diachronic regularities are not necessarily self-evident. Although the digital traces are temporally ordered, issues like simultaneity, recurrence, and recursion can render the temporal interpretation of patterns ambiguous. Therefore, it is not only the terminology in a lexicon that offers guidance, but also the time-ordered relationships implied or previously established among the concepts. These commonly accepted relationships can serve as a starting point for temporal analysis.

In manual grounded theory, axial coding aims to relate concepts to each other in a way that supports the goal of the research—which typically involves identifying explanations of cause and effect or process explanations. Selective coding then aims to integrate those explanations in relation to one core category representing the phenomenon studied, thereby producing an integrated theoretical scheme. In computational analysis, establishing diachronic relationships is achieved with some sort of inductive model. One form of inductive model is referred to as a "structural model,"[4] which relates concepts into quantitative laws in the form of rules, equations, and models (Rose and Langley 1986). Various computational techniques exist to favor parsimonious explanations and uncover causal relationships from data (Pearl 2011), and resulting structural models are refined and validated by researchers (Saito and Langley 2007). A process model, another form of inductive model, focuses on the time-dependent relationships among concepts rather than assuming associations are stable over time. Concepts are often treated as states rather than variables, and ordering rather than correlation is used to relate them (Mohr 1982). Several techniques are available for grounded process theorizing from longitudinal data (Langley 1999, Van De Ven and Poole 1995). For example, temporal bracketing—one common technique—is used to distinguish different phases over which the phenomenon of interest unfolds and to analyze how actions of one phase lead to changes in the context that will affect action in subsequent

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phases (Langley 1999). Both computational and manual grounded theory analysis require a process of sensemaking; data analysis is ultimately a cognitive human process (Grolemund and Wickham 2014).

While longitudinal trace data spanning years and decades are easily obtained, they are missing the wider context that shapes relationships. Reconstructing context and relating it to emergent theory from data is one challenge given the overwhelming volume of data (Levina and Vaast 2016). Computational techniques such as process induction and process mining offer to extract temporal relationships such as ordering and sequencing from trace data (Bridewell et al. 2008, Günther and Van Der Aalst 2007). For example, Lindberg et al. (2016) use process modeling to gain an inductive understanding of how developers in open-source communities resolve software code interdependencies over time. Similarly, recent advances in social network analysis allow for understanding generative mechanisms that lead to a sequence of events based on past patterns of events (Butts 2008, Quintane et al. 2014). By focusing on the temporal dimension, these techniques extend knowledge established by structural models.

All aspects of computationally intensive theory development emerge throughout the research project. The researcher acts as a sort of detective that finds a lead in the data and then pursues that lead, looking at a variety of data sources using a variety of methodologies to construct valid theoretical propositions, while drawing on the lexicon of an existing community of researchers to move the understanding of those researchers forward. This approach is not entirely new, but it has been our goal to make it explicit in a general way for a general information systems audience.

5. Illustrations

We draw on three recent studies in information systems to illustrate the applicability of this approach to computationally intensive theory development (Miranda et al. 2015, Lindberg et al. 2016, Vaast et al. 2017). See Table 2 for a summary, followed by brief descriptions of each study.

Miranda et al. (2015) developed theory on how different facets of organizing visions influence the diffusion of IT innovations in companies. To do so, they applied supervised content analysis, network visualization, and statistical analysis combined with traditional content analysis. Their research question was shaped by authors' interest in institutional theory and unpacking institutional mechanisms at play in organizations. The study involved two inductive stages: (1) extraction of mental schema and hierarchical structure of organizing visions from archival documents and (2) exploration of facets of organizing visions in the diffusion of IT innovations. Throughout the process, the researchers iterated across multiple rounds of sampling and analysis. The initial sample involved 46 of the Fortune 50 firms. The sample included text from social media, product descriptions, and other media outlets. Collectively, this resulted in 2,414 text documents. In a second stage, the authors deliberately refined the sample to focus on a longitudinal panel of 1,183 initiatives that the researchers uncovered through manual analysis of the texts. Initial manual coding was subsequently automated through content analysis of the texts for the presence of six principles—civic, domestic, industrial, inspiration, market, and renown—drawn from the "orders of worth" lexicon. The six principles served as dimensions of the texts from which the authors sought to extract schemas of organizing visions. They used relational class analysis to identify four clusters, which they validated and labeled as follows: Efficiency-Engineer, Brand-Promoter, Good-Citizen, and Master-of-Ceremonies. After identifying the four schemas, the authors continued to investigate how the different visions affected diffusion with the sample of the 1,183 initiatives. They characterized differences in the schemas with four facets—coherence, continuity, clarity, and diversity—by considering the schema variation over time. They then correlated the number of initiatives representing diffusion with the four facets. Visually examining the correlation scatter plots, the authors found that some of these relationships were linear while others were quadratic. From this analysis, they theorized that organizing visions are hierarchies of schemas and that different facets in these hierarchies differentially drive the diffusion of IT innovation.

Lindberg et al. (2016) explored an open-source software community (Rubinius) to understand how community developers coordinate complex work in ways that go beyond arms-length coordination mechanisms. They mixed sequence and statistical analysis with manual coding and visual interpretation of data to develop a process theory for coordinating around unresolved interdependencies in such communities. Initially they sampled 686 pull requests across 12 months of an open-source software project that included 3,704 activities. Initial rounds of computational analysis involved sequences and sequence covariates of time-stamped activities labeled in the software development platform (GitHub activities: assigned, closed, commented, mentioned, merged, opened, referenced, reopened, reviewed). This analysis led to initial identification of pull requests that were variably complex. In referencing the lexicon from coordination and organizational routines theory applied to software development, they characterized this complexity in terms of "unresolved" developer and development interdependencies, and activity and order variation in the routines. They used combinations of regression and visual inspection to

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Table 2. Illustrations of Computationally Intensive Theory Development (Manual Aspects in Brackets)

	Miranda et al. (2015)	Lindberg et al. (2016)	Vaast et al. (2017)
Sampling[a]	[Decision to explore corporate use of social media and IT innovation] —Round 1: 2,414 texts [Decision to filter data to focus on early adopter firms] —Round 2: 1,183 initiatives panel	[Decision to explore Rubinius project] —Round 1: 686 pull requests with 3,707 activities [Manual examination of text attached to sequences] —Round 2: 432 text excerpts	[Decision to explore microblogging around oil spill] —Round 1: 23,000 tweets [Decision to focus data collection on three connective action episodes] —Round 2: 1882 tweets
Synchronic analysis	[Manual coding: civic, domestic, industrial, inspiration, market, and renown] Automated coding [Sensemaking of cluster analysis] Categories of vision: Efficiency-Engineer, Brand-Promoter, Good-Citizen, and Master-of-Ceremonies [Sensemaking of network and temporal representation of data] Facets: coherence, continuity, clarity, and diversity	Initial (activity) codes: assigned, closed, commented, mentioned, merged, opened, referenced, reopened, reviewed [Referring to routine literature lexicon] Constructs: developer and development interdependencies, order and activity variation [Manual coding] Qualitative codes: Diagnosing, Causal theorizing, Asking for clarification, Clarification, Teaching, Adding features, Increasing code clarity, Increasing code functionality, Asking for tests, Providing tests, Asking for documentation, Providing documentation [Axial coding] Qualitative categories: knowledge integration; direct implementation	Episodes: Boycott BP, Stop the Drill, Hair and Fur [Sensemaking of cluster analysis] Group clusters: advocates, supporters, and amplifiers [Sensemaking of time series and motif analysis] Enacted role characteristics: roles, frequency, intensity, pattern of feature use, actions, reciprocal interdependence
Lexical framing	Organizing vision and diffusion of innovation theory; orders of worth	Coordination and organizational routines theory; coordination in online communities	Organizational interdependency theory; affordance theory
Diachronic analysis	Facets of different visions associated with differential diffusion	Process theory of coordination around unresolved interdependencies through direct implementation or knowledge integration	Theory of the role of artifacts in different connective actions
Benefit of combining manual and computational analyses	Automation	Complementarity	Creation of alternative representations

[a]Note that in each of the illustrations, sampling was an iterative process that was more complex than shown here. We briefly discuss this point when summarizing the three studies at the end of Section 5.

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identify associations among types of interdependencies and routine variation. The final elements of their theory generation involved manual, qualitative analysis of a sample of 432 text excerpts from these complex pull requests. They qualitatively coded the second data set using a traditional GTM approach through multiple rounds of coding (final codes: diagnosing, causal theorizing, asking for clarification, clarification, teaching, adding features, increasing code clarity, increasing code functionality, asking for tests, providing tests, asking for documentation, and providing documentation; final categories: knowledge integration and direct implementation). They concluded with a process theory of coordinating unresolved interdependencies in online communities through the mechanisms of knowledge integration and direct implementation.

Vaast et al. (2017, p. 1179) combined grounded theorizing with clustering, network motif analysis, and time series analysis to examine how social media use affords new forms of organizing and collective engagement. Their paper explores an oil spill in the Gulf of Mexico to understand new forms of collective engagement that they refer to as "connective action." Given this focus, the authors decided to sample data from the microblog service Twitter. The study began with an initial sample of 23,000 tweets related to the Deepwater Horizon incident in April 2010 to broadly gain insight into microblogging activity in the wake of a disaster. The choice of this crisis was deliberate—because of its magnitude, the crisis led to various forms of collective action. On Twitter, this was "the most microblogged issue in 2010" (Vaast et al. 2017, p. 1184). From a first round of manual open coding, three threads of communication they described as connective action episodes (CAEs; Boycott BP, Stop the Drill, Hair and Fur) emerged. This observation then informed subsequent sampling. A second round of sampling focused on extracting all tweets related to the CAEs through trackbacks of originally identified tweets. Based on this refined theoretical sample, they conducted a new round of manual open coding focusing on the similarities and differences between the CAEs that resulted in a number of role categories. They switched to automated taxonomy creation to identify role categories by performing a cluster analysis using the DBSCAN algorithm on their patterns of Twitter usage. They then looked at how members of these clusters participated in the three CAEs to contrast among episodes. The paper focuses on two types of associations: CAEs with actor categories, longitudinally with temporal analysis, and actor categories within CAEs, cross-sectionally using social network motif analysis. The temporal relationships are visualized with time-series plots to identify patterns. These patterns reflect interdependencies among actor categories. By characterizing the type of interdependence among actors in CAEs, the authors draw on different organizational theories of coordination and interdependency. Integrating the new lexicon with the theory of affordances, they introduce a theory of the role of connective affordances in the context of connective action.

The combination of manual and computational approaches allowed the authors of the three papers to go beyond what could have been achieved using only traditional methods. While the data collected in Miranda et al. (2015) lend themselves to manual coding, identifying and tracing organization visions over a long period of time is extremely challenging and perhaps not feasible. The supervised content analysis approach allowed the researchers to create categories based on their pre-theoretical understanding of the phenomenon and their exploratory analysis of a subsample. The benefit of the computational content analysis was to automate the lengthy and tedious process of manually coding the six-year data of 50 companies. In Lindberg et al. (2016), the collected sequences from the open-source repository have textual elements, including descriptions and comments by software developers. While sequence data lend themselves naturally to computational analysis, textual elements are better understood with human sensemaking. The two methods are complementary. Finally, the exploratory manual analysis of collected data by Vaast et al. (2017) led them to focus on connective affordances of social media. Understanding the interconnections of a large number of people is a cognitively challenging task. The value of the computational methods was to complement manual sensemaking to create alternative representations to understand connective action at scale. The network motif analysis provided a visual summary for the researchers to interpret.

We verified these insights by communicating with all authors of these three papers to understand their own views of the challenges in applying computationally intensive approaches. The authors highlighted a number of challenges, the foremost of which involved identifying the appropriate reference lexicon. The authors used expressions such as "tying together the different visualizations with a coherent theoretical narrative" (Lindberg) and "connecting the data to a theoretical anchor in a way that made sense conceptually and that respected the collected data" (Vaast) to describe the key challenges. Furthermore, they pointed out that there is no straightforward, mechanistic way to enact these approaches. They are intensely iterative and creative processes that have no specific guidelines. Although the iteration between the sample and the analysis is often downplayed in the reporting, authors of all three studies indicated an intensely emergent process of sampling decisions and continuous analysis. As one author mentioned, "the method we adopted was fairly

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emergent and idiosyncratic, it was not easy for us to refer to established guidelines for mixed-methods research . . . they could only provide templates that did not fully fit what the study was doing" (Miranda). All authors indicated that the bulk of the visualizations used along the way to develop their stories never made it into the final versions of their papers. The authors pointed to open and innovative reviewers and editors who helped them to construct their stories in convincing ways. Finally, authors noted that things seem to be changing. They find that there is an increasing variety of tools (such as new R packages) and an increasing appreciation of computationally intensive approaches. As one author put it, "The tone in our community is increasingly one of accepting that computational tools will be important even to qualitative scholars and those focused on theory development" (Lindberg).

6. Discussion and Implications

> Everyone who publishes in professional journals in the social sciences knows that you are supposed to start your article with a theory, then make deductions from it, then test it, and then revise the theory. At least that is the policy that journal editors and textbooks routinely support. In practice, however, I believe that this policy encourages—in fact demands, premature theorizing and often leads to making up hypotheses after the fact—which is contrary to the intent of hypothetico–deductive method. (Locke 2007, p. 867)

In Locke's (2007) quote, he points out how the policy of presenting research in terms of constructing hypotheses and testing them often goes against the real work of theory generation and empirical analysis. Quite frequently, findings are developed inductively, and then researchers disingenuously reconstruct a hypothetico-deductive paper out of the patterns they found (Anonymous 2015). This is because there is a stigma associated with "fishing" in data for patterns that may simply be spurious correlations without theoretical explanations. Post hoc rationalizations that justify results after the fact are to be avoided—and for good reason (Garud 2015, Walsh 2014). In the age of computational social science and trace data, there should be a mechanism for this pretense to come to an end. How can researchers inductively generate theory from patterns they see in data without feeling the need to repackage their research in terms of hypothesis testing? There is an important place for inductive theory generation, but researchers must be honest about what they are doing (Garud 2015). The answer lies in a general approach that highlights the role of lexicons in the emergent process of analyzing multiple forms of data—including trace data—for the purpose of generating theory.

For decades, qualitative information systems researchers have understood that rigorous attention to empirical data via cycles of sensemaking can help generate novel theory. Using GTM, qualitative researchers have a legitimizing tradition to draw upon when explaining patterns they see in accordance with existing lexicons and proposing the resulting ideas in terms of theory generation (Walsh 2014). We have highlighted the relevance of lexical framing in the process of identifying both concepts and relationships between concepts, and thus in facilitating theory emergence. Glaser and Strauss (1967) led a revolution of sorts in social analysis. Through a program of intense attention to empirical data, they legitimized a way to generate novel theory that could revitalize a stale discourse. Some argue that organizational and information systems literatures may be stagnating (Davison 2010) or not reaching their potential (Grover 2013). Now, particularly given the opportunity that the data explosion provides, it is time to open up approaches for theory generation that are grounded in empirical data. At the same time, it is important to capitalize on the maturity and flexibility of GTM, and to encourage further methodological attention in this regard to get the most out of on the new opportunities proffered by the availability of trace data. Against this backdrop, we join with those calling for a broader grounded *paradigm* for theory development based on the key features of grounded theory (Walsh et al. 2015). This paradigm is characterized by two key elements: the "grounded" and the "theory." *Grounded* refers to the intense attention to empirical data, comprised of rounds of sampling and analysis using a variety of qualitative, quantitative, and computational techniques. *Theory* refers to the patterns of associations that emerge from this analysis as researchers construct their understandings of the phenomena by drawing on and extending the lexicon of a community of researchers. In this paper, we have sketched a broad approach for information systems researchers dealing with any type of data using computationally intensive approaches to theory development.

It was recently stated that information mining and traditional theory building are indeed complementary, interrelated methods (Dhar 2013, Gopal et al. 2011). Data and knowledge mining methods by themselves, however, do not move forward the understanding of a phenomenon. To move this understanding forward, we need to theorize and explain the patterns of association that we identify. To make sense of patterns identified through computational methods, and to form appropriate mental models that can be used in the sensemaking process (Holland et al. 1986), the analyst requires a lexicon that is shared by a community of scholars (Habermas 2003). This lexicon can be taken from existent theoretical lexicons (such as the social network lexicon), which, in turn, serve as the pre-theoretical lexicons in the process of novel theorizing. Similarly,

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the patterns generated through computational analysis constitute pre-theoretical understanding in the form of synchronic regularities that can serve as a foundation for the development of novel theory. Our framework can thus be seen as an answer to the call made by Gopal et al. (2011, p. 370), who suggests that "researchers may develop an iterative approach that uses information mining outcomes as inputs into the theory construction and validation processes." Overall, theory developed from a combination of techniques can be more robust than theory generated from a single qualitative data set, as researchers triangulate and cycle through different approaches (Van de Ven 2007). The general approach to computationally intensive theory development accommodates different combinations of manual and automated activities. It draws attention to the opportunity afforded by the widespread abundance of trace data, and finds that the interplay of manual and computational techniques together can drive novel theorizing and is entirely consistent with GTM, but is also open to other forms of computationally intensive inductive analysis. This approach is just a start, and more work is needed to flesh it out. Other should push the grounded paradigm further.

The information systems field is particularly well positioned to lead a methodological revolution in computationally intensive social research (Agarwal and Dhar 2014). As a discipline, we investigate those phenomena that have made the trace data revolution possible in the first place. Our discipline is devoted to investigating complex sociotechnical settings that require us to make sense of large amounts of data that pertain to the interaction of "the social" and "the technical" (Orlikowski 2007). Furthermore, there is a very real need to develop novel and accurate theory grounded in large amounts of data instead of continually "working" existing theories (Legewie and Schervier-Legewie 2004), as we are challenged to further develop our intellectual core. The importance of new methodological approaches cannot be underestimated. Some of the most important, innovative, Nobel Prize–winning findings owe themselves to methodological advancements (Greenwald 2012). If one were to compare this trace data opportunity in social science to physics, it would be "as if every physicist had a supercollider dropped into his or her backyard" (Davis 2010, p. 696), and the field of information systems is poised to contribute.

7. Conclusion

> On the one hand, there isn't a paradigm for data scientists to easily publish their inductive data discoveries. These, often highly insightful, findings either go unpublished or are turned into hypotheses followed by testing to suit mainstream publication requirements. On the other hand, grounded theory scholars are increasingly encountering large digitized data archives that cannot be reasonably analyzed with qualitative methods alone. Thus we would all benefit if we start including inductive data scientists into the grounded theory research community and start using some of the advanced analytical techniques available today.
>
> (Levina in Walsh et al. 2015, p. 11)

In her quote above, Levina points to the opportunity presented by the abundance of trace data and argues for incorporating computational analyses in our empirically grounded theory development efforts. In this research commentary, we inquire into how the lessons learned from GTM can be used to build theory from trace data, thereby building a general framework for this approach. Specifically, we highlight the importance of a lexicon in this process. It is perhaps noteworthy that the development of our approach itself indeed has grounded components, too. We looked to what, at first blush, may be described as polar extreme approaches to theory development—the automated CTD tradition and the GTM approach. In relating the two, we find that there is quite a bit of similarity at a general level of abstraction, and we develop a general approach based on this similarity.

Armed with this general approach, we encourage researchers to act like detectives when looking to generate theory. It is important to note that this is not a methodology, per se, but a general approach whereby researchers creatively use qualitative approaches as needed, but do so in conjunction with a variety of computational techniques—ever employing new techniques as they come online—to triangulate and validate insights and conjectures, resulting in potentially more robust and creative theorizing. Their detective work, however, cannot ignore the cumulative knowledge of the community of scientists, and it is critical to highlight the role of lexicons as the source of and destination for that knowledge.

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Endnotes

[1] Note that the definition of "theory" is a contested issue (see DiMaggio 1995, Sutton and Staw 1995, Weick 1995), but to conceive of theory in terms of general statements about the relationship among concepts is a commonly accepted view (Jaccard and Jacoby 2010).

[2] Glaser (1978, 1992), for instance, distinguishes the stages of open, selective, and theoretical coding.

[3] Note that in axial coding, the analyst also starts to identify whether the categories are indeed conditions or consequences, and the lines between synchronic and diachronic analysis blur. We illustrate our model by using GTM terminology borrowed from Strauss and Corbin (1990, 1998). In Glasarian GTM, synchronic analysis would primarily comprise open and selective coding, where the former

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would identify first categories and the latter groups categories further (Glaser 1978).

[4]Note that the term "structural model" has a somewhat different meaning in different fields. Here we use it to describe abstract models of stable relationships among variables.

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